Q1

Query → Select t.tech_name, t.phone_number
from Technicians as t join Examine as e
on t.SSN = e.SSN
where e.date in ('10/27/2021', '10/28/2021');

a) $\pi_{t.phone_number,\ t.tech_name}(\overset{rename}{\rho_{t}}(Technicians) \bowtie_{t.SSN} (\sigma_{e.date\ in('10/27/2021', '10/28/2021')}(\overset{rename}{\rho_{e}}(Examine))))$

b) R.A-Tree

$\pi_{tech_name,\ phone_number}$

↑

$\sigma_{date\ in('10/27/2021', '10/28/2021')}$;

↑

$\bowtie_{t.SSN = e.SSN}$

↗ ↖

Examine Technicians

After optimization

$\pi_{tech_name,\ phone_number}$

↑

$\bowtie_{t.SSN = e.SSN}$

↗ ↖

π_{SSN} π_{SSN}

↑ Technicians

$\sigma_{e.date\ in}$

$('10/27/2021', '10/28/2021')$

↑

Examine

c)

$\bowtie$

Examine Technicians

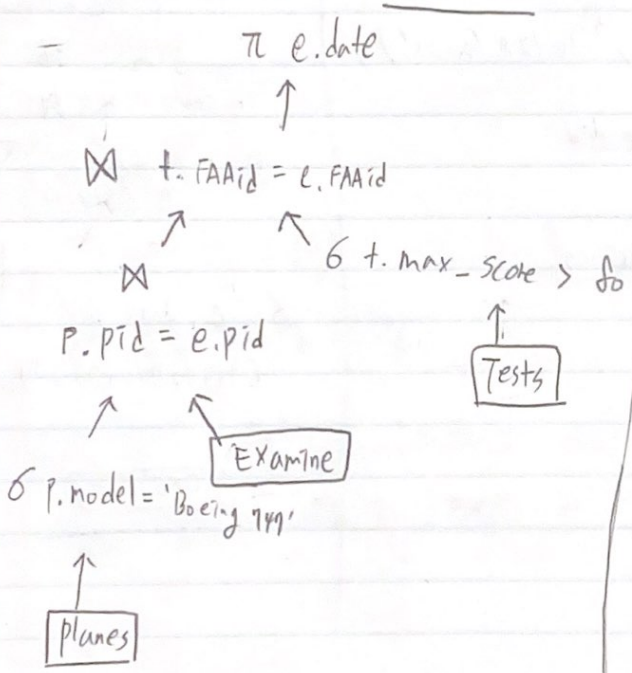
Examines Planes

Q2 Query : Select e.date
from Examine as e join planes as p
on p.pid = e.pid join Tests as t
on t.FAAid = e.FAAid
where p.model is 'Boeing 747' and t.max_score > 80
group by e.date
having count(*) > 0;

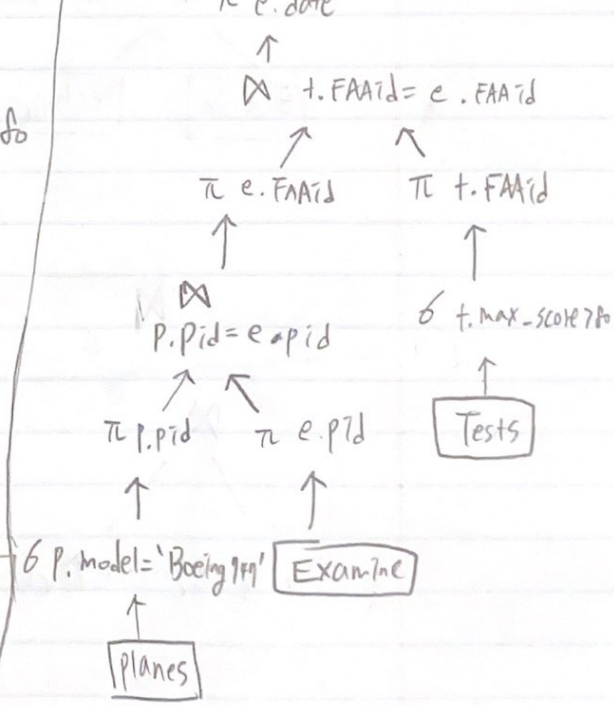
a) $\pi_{e.date}(\sigma_{c>0}(_{e.date}\mathcal{G}_{count\,c}(\overset{\text{rename}}{\rho_e}(Examine) \bowtie_{e.pid} (\sigma_{p.model=\text{'Boeing 747'}}(\underset{\text{rename}}{\rho_p}(Planes))) \bowtie_{t.FAAid=e.FAAid} (\sigma_{t.max_score>80}(\underset{\text{rename}}{\rho_t}(Tests))))))$

R.A-Tree

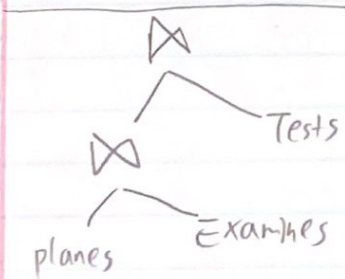
b)



After optimization



c)

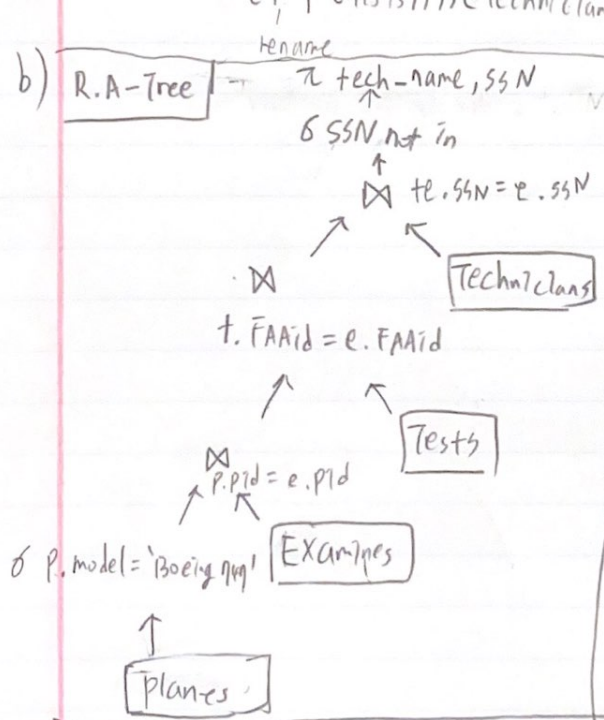


Q3 Query :

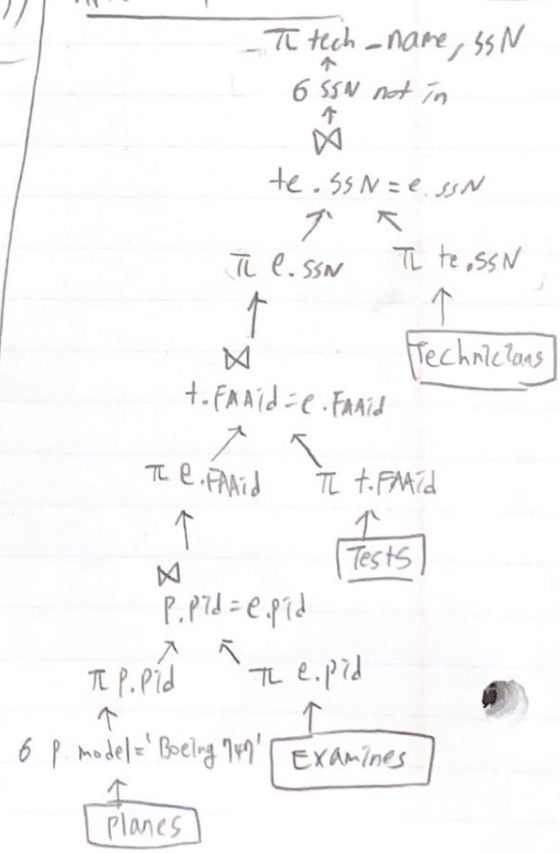
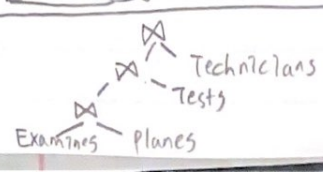
```
Select t.tech_name, t.SSN
from Technicians as t
where t.SSN not in (
    Select te.SSN
    from Technicians as te Join examine
    on te.SSN = e.SSN Join Planes as P
    On t.FAAid = e.FAAid Join Tests as t
    on P.Pid = e.pid
    where P.model is 'Boeing 747'
    );
```

a) $\pi_{tech_name, SSN}(\sigma_{SSN \text{ not in } (\pi_{te.SSN}(\overset{\text{rename}}{\rho_{te}}(Technicians)) \bowtie te.SSN(\underset{\text{rename}}{\rho_{e}}(Examine)) \bowtie P.pid = e.pid(\sigma_{P.model = 'Boeing\ 747'}(\underset{\text{rename}}{\rho_{P}}(Planes))) \bowtie t.FAAid(\underset{\text{rename}}{\rho_{t}}(Tests)))}(Technicians))$

b) R.A-Tree



After optimization



c)